Mid-Portion

BT601 - Virology:

1) Genomic Properties of Togavirus: (48) ★ ★ 32

Genome:- Linear, non-segmented, single-stranded-positive-sense RNA. ★ Size:- 10,000-12,000 nucleotides

★ Vision:- Enveloped, spherical (65-70nm diameter)

★ Capsid:- Icosahedral, composed of 240 monomers (T=4)

Protein and Glycoprotein encoded by Orthomyxovirus:

2) Orthomyxoviride Proteins: ★ 33

1. Hemagglutinin (HA): virus attachment and entry.
2. Neuramindase (NA): Release of new virions
3. Matrix protein (M1): Viral envelop and RNA linking
4. Nucleoprotein (NP): Viral RNA replication & transcription
5. RNA polymerase (PB1, PB2, PA): viral RNA synthesis.

3) Host Range of Virus: ★ (49) ★ 34

Viral host range refers to the specific cells and organism, a virus can infect.

Determining Factors:

*Specific attachment sites on host cells

*Availability of host cellular factors.

Examples:- Plants, vertebrates, invertebrates, Protists, Bacteria, fungi.

4) Bunya virus genome segments & encoded proteins (50) ★ ★ 35

RNA dependent (RdRp)

1- Large: RNA polymerase

2- Medium: Glycoproteins (Gn, Gc) and NSM m (Non-structural protein) medium

3- Small: Nucleocapsid protein (N) and NSs (Non-structural protein) small

Bunya virus consist of three RNA segments called large, medium, small that encode four structural proteins, as well as one or two non-structural protein

(5) SEM and TEM * (51)

SEM	TEM
• Scanned Electron Microscope	Transmission Electron Microscope
• Electron beam scans specimen surface	Focused electron beam passes through spec.
• Primary e^{-1} produce secondary e^{-1}	Beam focused by electromagnetic condenser lens.
• Secondary e^{-1} create image on screen or plate	Producing detailed internal image.

(6) * Positive Sense RNA Names: * 10 plants viruses: (52)

- Bromoviridae
- Cucumavirus
- Bromovirus
- ilavirus
- Alfamovirus
- Ourmiavirus
- Tobamovirus
- Hordeivirus
- Pecluvirus
- Furovirus
- Pomovirus
- Bemivirus

(7) * Effects of Viral attack on host cells: * (53)

1. Cell damage/destruction
2. Tissue damage/inflammation
3. Organ dysfunction/cell death

(8) Names monochromatic light sources used in microscope: *

- Xenon arc lamps
- Mercury vapor lamps
- Super Continuum source
- High power LED

(9) * In-Vitro Genetic Techniques: (54) *

- Physical Analysis
- Nucleic Acid Isolation
- DNA sequencing (endonucleases, vectors)
- Electron Microscope
- RNA sequence Analysis (reverse transcriptase)
- PCR
- Molecular cloning
- Gel Electrophoresis.
- Hybridization of nucleotide sequences
- Gene editing (CRISPR/Cas9).

Phage Symmetry Types ★ (56)

(10) Comparison of 3 types of viral Symmetry.

1- Helical Symmetry (Rod-shaped) (Cylindrical.) with spiral arrangement

2- Icosahedral Symmetry (Spherical viruses) Polyhedral shape with 20 triangular faces

3- Complex Symmetry (Combination of helical & Icosahedral). Poxvirus.
→ variety of shapes and structure.

(11) Factors for viral Transplant:

1- Immune system suppression

2- Species-to-species transmission (e.g. primate to human).

(12) General Characteristics of Viruses: ★ (57) ★ 41

★ Microscopic infectious agents ★ Replicate inside host cells

★ Simple structure (DNA/RNA + capsid) ★ Non-living

★ Highly specific ★ Infect various organisms ★ Causes diseases

★ Transmitted through various routes

(13) Replication of Coronavirus: ★ (58)

Coronavirus replication involves a complex series of steps.

1- Attachment (S protein receptors) 2- Endocytosis (virus entry)

3- Membrane fusion (virus release) 4- Genome release

5- Polyprotein synthesis and cleavage 6- dsRNA genome synthesis

7- Transcription (mRNA/genome synthesis) 8- Structural Protein synthesis

9- Assembly & Budding 10- Release of new virions (exocytosis)

(14) M-13 Phage: ★ ★ (59) 42

1- Simple genome (6.4kb) 2- Single stranded, +ve sense, circular DNA

3- Encodes 10 genes 4- Icosahedral symmetry 5- Expandable capsid.

(15) Viral Tropism:

Ability of a virus to infect specific cells, tissues or hosts.

Example: Virus infecting rabbit macrophages but not other cells.

(16) Retrovirus Replication: * (60) *

- Reverse transcription (RT) produces DNA from RNA.
- DNA integrated into host genome (provirus)
- Host cell transcribes & translates viral genes.
- Proteins produced for new virus assembly.
- Infection persists indefinitely.

(17) * Retrovirus genomic Properties: OR Genetics of Retrovirus (61)

* Single-stranded RNA * Positive sense * Enveloped
* Three genes: GAG, ENV, POL * Icosahedral shape.

(18) Bunyavirus: (Capsule protein p24) (Envelop Glycoprotein gp41 - gp120) (Reverse transcriptase, Integrase activity)

⇒ Clinical Manisfestation: Fever, rash, hemorrhage, hepatitis
⇒ Structure: Spherical, enveloped particles (90-100nm),
3 segments of single-stranded RNA.
⇒ Classification: 5 genera: i- Orthobunyavirus ii- Hantavirus
iii- Nairovirus iv- Phleovirus v- Tosporivus

(19) How Edward Jenner made vaccine? * (62)

* Edward Jenner (1796) introduced the term Vaccination
* Used cowpox to confer immunity to smallpox.
* First successful trial of vaccine in 1796
* Saved million of lives worldwide.

(20) * Theories of Virus Evolution: * (63)

1- Co-evolved with other life forms
2- Originated from cellular nucleic acids.
3- Regressive theory (lost genetic information)

(21) Subfamilies of Herpesviridae: ★ (64) ★ 47

1- Alpha herpes virinae (Herpes simplex virus)
2- Beta herpes virinae (Cyto megalovirus)
3- Gamma herpes virinae (lymphoproliferative virus)

(22) Lifecycle of Coronavirus:

1- Attachment 2- Penetration 3- Uncoating
4- Gene expression & ~~Assembly~~ Replication: SER
5- Replication 6- Release

(23) Ways of Reversion: ★

• True reversion (back mutation) • Intragenic reversion (compensatory mutation) • Extragenic reversion (supressor mutation)

(24) Neutralizing Test: ★ It is a serological test that (bind to viral particle) 4% ★ (65)
Detects viral neutralizing antibodies that block viral infectivity and prevent from viral infection.

(25) Baltimore Classification: ★ 1971 ⇒ Type of nucleic acid ⇒ Mode of replication ⇒ Host organism ⇒ Diseases caused ★
Categorizes viruses based on nucleic acid genome and replication strategy. (66) 49

1- ds DNA viruses 2- ss DNA viruses 3- ds RNA viruses
4- +ve sense ssRNA viruses 5- -ve sense ssRNA viruses
6- RNA viruses that ↳ reverse transcoibe 7- DNA virus that reverse transcribe.

(26) Properties of Picornavirus: ★ ★ (67) Family: Picornaviridae +ve sense RNA (~7.5 kb) 50

• Small (22-30nm), non-enveloped • Single-stranded RNA
• Icosahedral symmetry • Composed of 60 protein subunits (VP1-VP4)
* Insensitive to ether, chloroform, or non-ionic detergents.

(27) Which type of Centrifugation used for isolation of virus?

Ultracentrifugation are used for the isolation of virus.

- They operate at high speeds (upto 100,000 rpm) and
- can separate viral particles based on size, density and buoyant density.

(28) Contribution of Iwanowski and Beijerinck:

* Dimitri Iwanowski (1892): showed disease transmission in tobacco plants using filtered extracts.
* Martinus Beijerinck (1898): confirmed Iwanowski's results, developed the concept of contagium vivum fluidum (soluble living germ).

(29) Latent Viruses:

"Pathogenic viruses that lie dormant (or latent) within host cells."

Example: HSV, Adenovirus, SSPE (Subacute sclerosis panencephalitis)

(31) Function of Condenser in Light Microscope:

* Focuses light rays onto the specimen under observation.

(30) Principle of Flourescence Antibody Technique (FAT)

- Antigen (virus protein) present in tissues.
- Flourescent-labelled antibodies bind to specific antigen.
- Bound antibody-antigen complex visualized using flourescence microscope.

RNA virus proofreading: Yes, RNA viruses have proof-reading, but they still show higher mutation rates than DNA viruses.

(32) Types of Mutation and Mutagens:

In-Vitro: Chemically modify nucleic acids (e.g., nitrous acid, hydroxylamine)

In-Vivo: Requires replicating nucleic acids (eg 5-bromouracil, UV irradiation)

(33) Types of Latency: * 53

1- Episomal Latency: • Viral genes in cytoplasm/nucleus,
• Vulnerable to immune system (e.g chicken pox, Herpes Simplex)

2- Provival Latency: Virus genome integrates into host DNA (e.g HIV)

3- Maintaining Latency: Continues infection, evades immune system.

(34) Agglutination and Precipitation Test: * (69)

- Agglutination: clumping of antigens with antibodies
- Precipitation: Soluble antigens bind with antibodies, forming an insoluble precipitate.

Principle: Viruses are sensitized by specific antibodies, forming aggregates

Applications: • Detecting antibodies if virus is known.
• Detecting unknown virus if known virus serum is available.

(35) Inoculation Methods:

I/E (Intraembryonic): Inoculation into embryonic tissues

I/C (Intracerebral): Inoculation into the brain

I/V (Intravenous): Inoculation into a vein.

(36) Dyes used in Transmission electron Microscope: * 54

1-Uranyl acetate 2-Lead citrate 3-Osmium tetroxide
3-Tungsten 5-Uranium (stains)

(37) Microscope: An instrument to see small objects (70) 58

Microscopy: The science of investigating small objects

Microscopic: Invisible to the eye unless aided by a microscope.

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Parts of Microscope and Functions

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1. Ocular lens: Magnifies image (10X)
Arm: Connect eyepiece to base
Stage: Holds specimen slide
Stage clips: Secure slide in place.
Objective lenses: Magnify image (4X, 10X, 40X, 100X)
Condenser: Focuses light on specimen.
Diaphram: controls light aperture.
Illumination / Light source: Provides light.
~~Coarse~~ Fine adjustment knob: Fine-tunes focus for individual vision
Coarse adjustment knob: Moves stage up/down for focusing
Base: Provide support for microscope.

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(1) Reason behind spontaneous mutations:

1- Mechanism of genome replication

2- Error-prone RNA-dependent RNA polymerase.

(2) Mode of Transmission of Arenaviruss

* Inhalation of virus from rodent urine, dropping, or nesting materials

* Contact with contaminated environment during cleaning or other activities.

(3) Types of Complementation:

1- Allelic (Intragenic): Different mutants with defects in the same protein.

2- Non-Allelic (Intergenic): Mutants with defects in different genes.

(4) Three disease cause by Peribunyaviridae:

1- Rift Valley fever 2- La Crosse encephalitis

3- California encephalitis

(5) Functions of Capsids:

• Protects genome • Prevents digestion by cellular enzyme

• Introduces genome into host • Binds to cell surface

(6) Apoptosis:

Regulated, programmed cell death with controlled cellular destruction.

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(7) 2 ✱ PrPres: Protease-resistant, structurally altered PrPc isoform, converted into in vitro

PrPSc: Infectious isoform, converts PrPc to misfolded form PrPSc, causes prion disease. host cell? also same

3 (8) How Viruses infect the Human Body?

⇒ Viruses infect humans by; Cell damage/destruction Tissue damage/Inflammation organ dysfuction/ cell death

1-Attaching to host cells 2-Replicating

3-Producing new virus particles

⇒ Causing damage through; & inflammation.

•Direct cell damage •Triggering immune response

4 (9) Prions: ✱

*Abnormal, mutated protein * No DNA or RNA

*Causes lethal diseases (e.g., neurodegenerative diseases)

*Misfolded proteins, infectious

*Characterized by fatal disease in animals & humans

(10) Structure of Bacteriophages:

- Head: contains genetic material
- Tail: attaches and injects genetic material into host bacteria.
- Tail fibers: (optional): help locate and bind to host cells.

5 (11) Centrifugal Technique: ✱

"Separates materials by density using centrifugal force."

Principle: Denser components move down lighter up.

Types: Differential Centrifugation • Density Gradient Centrifugation.

Mechanism of plant resistance to virus:
- R gene mediated resistances
- RNA Silencing.

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(12) Three Filteration Methods: ★ 3 porosities: V(Coarse): filteration preliminary
N(normal): retains bacteria, allow viruses to pass
W(Wein): retain bacteria & large viruses to filter out bacteria
6 ★ => Berkefeld Filter: Uses diatomaceous earth and particles to remove bacteria
=> Mandler Filter: Uses porcelain filter and contamination.
=> Pasteur-Chamberland Filter: Uses an unglazed porcelain to remove bacteria & particles, used for sterilizing liquids

(13) Purpose of strand/Multiple displacement Amplification: is to amplify minute amounts of DNA to a reasonable quantity for genomic analysis.

(14) Scientist's Role in Tobacco Mosaic Virus (TMV): Two key scientists played a crucial role in understanding TMV:
1- Wendell Stanley: Discovered TMV composition (protein & nucleotide) and isolated the virus.
2- Martinus Beijernick: Developed the modern concept of viruses & coined the term "Contagium vivum fluidum".

7 (15) Viruses transmitted in plant mode: ★
Viruses transmitted in plants through;
1- Vector transmission (insects/animals) 5-Sap Transmission (TMV)
Pollen Transmission 2- Seed transmission (infected seed)
Mechanical Transmission 3- Grafting (infected scion) 4-Mechanical Transmission (infected sap, contaminated tools/machinery).

(16) Verticle transfer of viruses: refers to transmission from parent to offspring through; * Parental * During Birth * Breastfeeding
3 Examples: HIV, Zika Virus, Herpes simplex virus

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(17) Function of Antibiotics:

- kill/inhibit growth of bacterias fungi & microorganisms
- Disrupt essential cellular processes.
- Treat bacterial infections.

(18) Function of Trypsin.

- Breaks down proteins into peptides
- Aids digestion and nutrient absorption.
- Regulates protein synthesis/degradation.

(19) Latent Occult Virus infection

Pathogenic viruses lying dormant within host cells

Examples: 1-HSV (Herpes simplex virus) 2-Adenovirus 3-SSPE (Subacute Sclerosing Panencephalitis)

(20) Types of Pathogenesis in interaction of virus & host cells

- Inapparent • Apparent • Acute • Subacute
- Chronic • Latency

(21) Activation of apoptosis is done by virus:

1-Lytic granules release:

- Perforin (disrupts cell membrane) • Granzyme (affects cell process)

2. CTLs trigger apoptosis through

- Ligand binding • Apoptosis cascade initiation

Cytotoxic T-lymphocytes.

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(22) Phages Replicate through 2 cycles; *

Lytic Cycle	Lysogenic Cycle
*Attachment	*Temperate phage,
*Virulent phage	Integrates into host
*Kills host cells (lysis)	* genome (provirus)
*Releases new phages	* Transmitted to daughter cells.
	* Can initiate lytic cycle under stress

Attachment Entry Synthesis Assembly Release

(23) Advantages: rapid replication
Limitation: Host killed, preventing viral genome passage

Advantages: More viruses produced, passed to future generation
Limitations: slower replication, dependent on host cell replication.

(24) Roller Tube Cultures: *

"Grown in test tubes and rotated during incubation."

Procedures: • Tissue explants in test tubes
• Plasma/agar added • Tube rotated (8-12rpm) during incubation
• Tissue explants partially submersion in nutrient media

(25) Types of Interferons:

Type I:
1- Interferon alpha: maximal antiviral activity
2- Interferon Beta: intermediate antiviral activity

Type II:
Interferon Gamma: more lymphokines than antiviral

(26) Regulation of Interferon Expression:

1- Normally repressed 2- Viral infection removes expression
3- Interferon synthesis occurs.

(27) Clinical Uses of Interferon:

IFN-A: Treats viral infections & Cancer
IFN-BG: Immunostimulant

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(28) Evasion of Immune Response by Virusess.

Viruses evade the immune system by;

•Avoiding detection •Interfering with host processes

•Depleting immune cells •Latency.

(29) 12 Targets of viral evasion proteins: ✱

•Pattern recognition receptoors •Interferons

•Dendritic cells •T-cell receptoors •Antibiotics

(30) Prevention of Viral Infections:

1- Vaccines: Live, killed, or subunit vaccine provide protective immunity.

2- Antiviral drugs: Inhibit viral replication, e.g. acyclovir (genital herpes), Tamiflu (influenza).

(31) Paramyxo:

Para = beyond & Myxo = mucus

Paramyxo & orthomyxo were formely known as myxoviruses

(32) Mode of transmission of arena viruses:

Rodent-to-human transmission through inhalation of virus in urine, saliva, or droppings

(33) Vaccine: Biological preparation providing active acquired immunity.

Vaccination: Injection of killed/weakened organism producing immunity

Immunization: Process of becoming protected from a disease.

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(34) Why Vaccines: * Safe * Effective & Community
*Works with immune system *Protects individuals

(35) Types of Vaccines:
1- Live-attenuated vaccines: (e.g., MMR, Chickenpox) Weakened form of the germ
2- Inactivated vaccines: (e.g., flu, hepatitis A) killed version of the germs that cause a disease
Subunit, Recombinant, Polysaccharide, & Conjugate Vaccines:
3- Specific pieces of the virus (eg., hepatitis B, HPV)
* 4- Toxoid Vaccines: (e.g. diphtheria, tetanus) Use a toxin made by the virus that cause a disease are used to protect against

(36) * Virus Vectors:
1- SV40 Based vectors: Small virus with circular dsDNA
2- Vaccinia virus vector: Large dsDNA with enzymes for replication.
3- Epstein Barr Virus: Herpes virus with large genome, used for episomal cDNA construction

(37) Gene Therapy:
Experimental technique using genes to treat/prevent disease.
Vectors: Viral, Non-viral, Physical (electroporation, gene gun), Chemical (oligonucleotides, nanoparticles)
Hurdles: *Short-lived immune response
*Viral vectors problems *Multigene disorders
14 *Mutagenesis *Cost-effectiveness

(38) Properties of Ideal drug
Chemotherapy against viral infections: *
=> Ideal Drug: water soluble, stable, non-toxic
=> Approved antiviral drugs: Amantadine, Ribavirin, Acyclovir etc
=> Limitations: narrow spectrum, ineffective against latent virus, toxic side effects.

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(39) Virus Titration

Measures virus concentration

Types: • Infectivity titer • Compulation titer

Standards:

1- LD50/mL (lab mice mortality)

2- Infectivity titer (smallest amount of virus causing manifestation)

(40) 13 Bacteriophage Titration: *

Determines phage particles in stock procedures:

1- Dilution preparation 2- phage & bacteria on agar surface

3- Incubation 4- Lysis of phage interrupts bacterial layer.

Note: "Charging" pipettor tips minimizes particle loss.

(41) 14 Plaque Count in Bacteriophage Titration: *

Measures phage concentration via plaque formation

Phage + host cells → lysis → plaque (zone of lysed cells)

(42) Gene: A DNA segment encoding a single polypeptide

Inbon: Non-coding region of a viral genome.

Exon: coding region of a viral genome.

Lysogen: A bacterial cell with dormant phage DNA (prophage)

(43) 17 Viroid Transmission: *

1- Aphids 2- Cross-contamination

3- Mechanical damage 4- leaf contact.

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(44) Prion Structure: *

- Protein (PrP) with 209 amino acids
- One disulfide bond • Alpha-helical structure
- Molecular mass: 35-36 kDa

(45) Fluorescence:

ability of substances to absorb short wavelengths of light (ultraviolet) and give off light at a longer wavelength (visible): used in fluorescence microscopy.

(46) T-phages: • Head and tail called binal symmetry.

• Icosahedral head and helical tail symmetry.

(47) Phase growth curves steps: (4 steps)

• Inoculation • Eclipse • Burst • Burst size

19 * * Transport media is used when transfer the viral infection.

20 (48) Red Queen Hypothesis: *

⇒ Organisms must constantly adapt and evolve to survive.

⇒ Their environment and competitors also evolve.

21 (49) Membrane Technology Applications: *

• Sterilization • Clarification • Water purification

• Emulsion separation • Reverse osmosis • Solid-liquid separation.

(50) 22 * Magnification = objective lens × Eyepiece lens

• Ratio of observed size to actual size (with unaided eye)

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(51) Factors affecting Virus titration: ★

1- Incubation time: Precise period for virus's reproduction

2- Virus type: plant/animal/Specific cell line

3- Conditions/media: Optimum for virus growth

4- Contamination: bacterial, fungal, procedural errors.

(52) Nucleic Acid Hybridization: ★

A technique where single-stranded DNA/RNA from complexes (hybrids) with complementary sequences

(53) Centrifugation Types: ★

1- Differential Centrifugation: separates by size/density in which particles are separated based on

2- Density gradient Centrifugation: buoyant density in a solution with continuous gradient of increasing or decreasing density.

(54) Hepatitis Delta Virus: ★

- Associated with hepatitis B virus
- 1.7kb circular, single-stranded RNA genome.
- Replicated by cellular RNA polymerase

(55) Benefits of phage therapy to other bacterial therapies: ★

Bacteriophages

- High specificity
- Harmless to host & gut flora
- High therapeutic index
- Few side effects
- Low effective dose.

★ Transport Media Methods: 1- UTM (Universal Transport Medium)

2- PBS/Glycerol transport Medium (containing NaCl, KCl, Na_2HPO_4 in d.w)

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(S7) ★ 29

Obligate Intracellular Parasites:

*Phages (bacterial viruses) * Plant viruses

Characteristics: •Both require host cell to replicate.

•Cannot reproduce outside host cell

•Depend on host cell's molecular machinery.

(S8) ★ 30

Viral growth identified in cell culture:

*Plaque formation * Cytopathic Effects (CPE)

*Flörescent Antibody Technique *Nucleic Acid hybridization

*Electron Microscopy *Hemadsorption.

(S9) ★ 31

Functions of Viral Proteins:

1- Structural: Capsid / envelop formation

2- Enzymatic: replication, transcription, translation

3- Regulatory: (gene expression, host interaction

(S5) Prions differentiate with virus & Bacteria

Viruses	Bacteria
• Infectious particles	• Single-celled microorganisms
• Contain DNA/RNA	• Contain DNA & cellular structures
• Require host cell for replication	• Can be beneficial (gut flora) or pathogenic.
• Cause various diseases (common cold to HIV)	
• Example: Influenza, HIV	• Example: E.coli, Staphylococcus